

AP Calculus AB: Pre-Class Assessment

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Challenge Questions

Directions: Solve each of the following problems. Calculators are NOT permitted on this section.

1. Evaluate the following limit:

$$\lim_{h \rightarrow 0} \frac{\sin(\frac{\pi}{6} + h) - \frac{1}{2}}{h}$$

(A) 0 (B) $\frac{1}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) The limit does not exist.

2. Let f be the piecewise function defined below, where a and b are constants.

$$f(x) = \begin{cases} x^2 + ax, & \text{for } x \leq 2 \\ bx - 3, & \text{for } x > 2 \end{cases}$$

If f is differentiable at $x = 2$, what are the values of a and b ?

- (A) $a = 1, b = 7$
(B) $a = -2, b = -3$
(C) $a = 5, b = 17$
(D) $a = -1, b = 3$

3. Suppose g is a function such that $\lim_{x \rightarrow 4} \frac{g(x) - 7}{x - 4} = 3$. Which of the following statements MUST be true? I. $g(4) = 7$ II. $g'(4) = 3$ III. g is continuous at $x = 4$

(A) I only (B) II only (C) I and II only (D) I, II, and III

4. The function $h(x)$ is defined as $h(x) = |x - 3| + 2$. Which of the following statements is true about h at $x = 3$?

- (A) h is continuous but not differentiable at $x = 3$.
(B) h is differentiable but not continuous at $x = 3$.
(C) h is both continuous and differentiable at $x = 3$.
(D) h is neither continuous nor differentiable at $x = 3$.

5. If $\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x^2 - 4} = 5$, what is the value of $f'(2)$?

- (A) 5
(B) 10

(C) 20

(D) Cannot be determined from the given information.